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ASSIGNMENT: 1

Academic Misconduct is an offence under university regulations, and this involves:

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- **Unauthorised use of artificial intelligence** – where you use artificial intelligence tools to generate your assignment instead of completing it yourself and/or where you have not been given permission to use artificial intelligence tools by your module leader. Please complete the following declaration around you use of artificial intelligence tools in your assignment.

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Live Demo Link

<https://ydvbishal4869-sketch.github.io/demo/css-demo.html>

CSS

If HTML represents the bare bones and blueprint of a website, **CSS (Cascading Style Sheets)** is the wardrobe, styling, and aesthetic vibe. It's the design language that transforms a sterile structure into a beautiful, engaging digital experience, giving you total control over how a website looks and feels.

What Does CSS Actually Bring to the Table?

Without CSS, the internet would look like a giant pile of 1990s legal documents—just endless rows of black text, bright blue hyperlinks, and stark white backgrounds. CSS is the magic that prevents that. It allows you to:

- **Paint the Canvas:** You can curate the perfect color palette, choose elegant typography, and mix-and-match fonts to set the right mood.
- **Shape the Architecture:** Instead of everything stacking awkwardly in a single vertical line, CSS lets you position elements side-by-side, build clean navigation menus, and arrange content into gorgeous layouts.
- **Be a Digital Chameleon (Responsiveness):** It ensures a website scales effortlessly. Whether someone is viewing your page on a massive desktop monitor or a compact smartphone screen, CSS automatically adjusts the fit.
- **Bring Static Pages to Life:** You can inject subtle animations, smooth fading transitions, and interactive hover effects that make the site feel responsive to human touch.

Why is it Called "Cascading"?

The word **Cascading** sounds fancy, but it actually describes a brilliant tie-breaking system. It represents the hierarchy CSS uses to settle arguments when formatting rules clash.

Imagine you accidentally give a single paragraph two conflicting instructions—one line says the text should be emerald green, but a few lines later, another rule says it should be midnight blue. Which one wins?

CSS solves this by "cascading" down a waterfall of priority rules based on two main things:

- **Specificity (The Target Factor):** A highly specific, targeted rule will always beat out a vague, generic one.
- **Order of Appearance:** If the conflicting rules are equally specific, CSS simply follows the latest instruction, meaning the rule written further down the page overrides the one above it.

Anatomy of a CSS Rule

CSS works by selecting an HTML element and then declaring what it should look like.

Here is a basic example:

CSS

```
/* Reset */ *{  
margin:0; padding:0;  
box-sizing:border-box;  
}
```

```
/* Body */
```

```
body{  
    font-family:Arial, sans-serif;  
    min-height:100vh;  
  
    background:linear-gradient(  
        135deg,  
        #667eea,  
#764ba2,  
        #ff6b6b  
    );  
  
    background-size:400% 400%;  
  
    animation:bgMove 10s ease infinite;  
  
    display:flex; flex-  
direction:column; align-  
items:center;  
    justify-content:center;  
  
    gap:40px;  
}
```

```
/* Animated Background */
```

```
@keyframes bgMove{  
    0%{  
        background-position:0% 50%;  
    }  
    50%{  
        background-position:100% 50%;  
    }  
    100%{ background-  
position:0% 50%;
```

```
    }  
  }
```

```
/* Title */  
.title{  color:white;  font-  
size:3rem;  text-  
transform:uppercase;  
letter-spacing:4px;
```

```
  animation:fadeIn 2s ease;  
}
```

```
/* Grid Container */ .container{  
  display:grid;  grid-template-  
columns:repeat(3,1fr);  
  gap:30px;  
padding:20px;  
}
```

```
/* Card */ .card{  
width:300px;  
padding:25px;
```

```
  background:white;
```

```
  border-radius:20px;
```

```
  box-shadow:    0 10px 25px  
  rgba(0,0,0,0.3);
```

```
  transition:all .4s ease;
```

```
  text-align:center;
```

```
  animation:slideUp 1s ease;  
}
```

```
.card:hover{  
transform:  
translateY(-15px)  
scale(1.05)  
  rotate(2deg);
```

```
  box-shadow:    0 20px 40px  
  rgba(0,0,0,0.4);
```

```
}

/* Glass Effect */
.glass{  background:rgba(255,255,255,.15);

        backdrop-filter:blur(15px);

        border:1px solid rgba(255,255,255,.2);

        color:white;
}

/* Button */ .btn{
margin-top:15px;

padding:12px 25px;

border:none;

border-radius:50px;

background:linear-gradient(
    45deg,
    #3498db,
    #8e44ad
);

color:white;

cursor:pointer;

transition:all .3s ease;
}

.btn:hover{
transform:scale(1.1);

background:linear-gradient(
    45deg,
    #ff6b6b,
    #feca57
);
}

/* Pulse Animation */
```

```
@keyframes pulse{
  0%{
    transform:scale(1);
  }

  50%{
    transform:scale(1.05);
  }

  100%{
    transform:scale(1);
  }
}

.pulse{  animation:pulse 2s
infinite;
}

/* Fade In */ @keyframes
fadeIn{
  from{
    opacity:0;
  } to{
    opacity:1;
  }
}

/* Slide Up */
@keyframes slideUp{
  from{
    opacity:0;
    transform:translateY(100px);
  }

  to{
    opacity:1;
    transform:translateY(0);
  }
}

/* Loader */
.loader{
width:60px;
height:60px;
```

```
border:6px solid rgba(255,255,255,.3);

border-top:6px solid white;

border-radius:50%;

animation:spin 1s linear infinite;
}
```

```
@keyframes spin{
  from{
    transform:rotate(0deg);
  }

  to{
    transform:rotate(360deg);
  }
}
```

```
/* Responsive */
@media(max-width:992px){

  .container{
    grid-template-columns:1fr;
  }

  .card{
width:90%;
  }

  .title{
    font-size:2rem;
  }
}
```

- **Selector (p):** Points to the HTML element you want to style (in this case, all paragraphs).
- **Property (color, font-size):** The feature you want to change.
- **Value (blue, 16px):** The settings you want to apply.
- **Declaration:** The property and value combined (e.g., color: blue;), always ending with a semicolon.

The Three Ways to Add CSS to HTML

You can apply CSS to your web projects in three different ways:

1. **External CSS (Best Practice):** You write your styles in a completely separate file (like style.css) and link it in the HTML <head>. This keeps your code clean and lets you style an entire website from one single file.
2. **Internal CSS:** You place the CSS rules inside a <style> tag directly within the HTML document's <head>. Good for single-page sites.
3. **Inline CSS:** You write the style directly inside the HTML element using the style attribute (e.g., <p style="color: blue;">). This is generally discouraged because it makes the code messy and hard to manage.